

First detection of anatoxin-a in human and animal dietary supplements containing cyanobacteria.

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Anatoxin-a is a potent neurotoxin produced by several species of cyanobacteria. This alkaloid may cause fatal intoxication to exposed organisms and this has raised concerns over the increasing popularity of food supplements containing cyanobacteria. These are being marketed with alleged health properties for animal and human consumption. These supplements most commonly contain the genera *Spirulina* (*Arthrospira*) and *Aphanizomenon* and their consumption represent a potential route for anatoxin-a exposure in cases where adequate quality control is not undertaken. In this work, several dietary supplements containing cyanobacteria from different commercial suppliers were evaluated for the presence of anatoxin-a by high performance liquid chromatography with fluorescence detection. Additionally, the presence of the previously derivatized anatoxin-a was confirmed by using Gas chromatography-mass spectrometry. A total of 39 samples were analysed in the study. Results showed that three of the samples (7.7%) contained anatoxin-a, at concentrations ranging from 2.50 to 33 microg g⁻¹. Quality control of cyanobacterial food supplements is required to avoid potential health effects in humans and animals.